

Formative R activities: Health Data Analysis and Hypothesis Testing

1. Objective

This explanatory analysis covers statistical activities conducted in R using the provided **Health Data** set. The dataset contains 210 observations and 17 variables, including age, sex, religion, occupation, income, and medical conditions such as blood pressure, peptic ulcer, and diabetes; pre- and post-test data; and admission and discharge data.

The activities focused on data exploration, data visualisation and various hypothesis testing in R. The R reporting script is available in the Appendix section.

2. Dataset Exploration

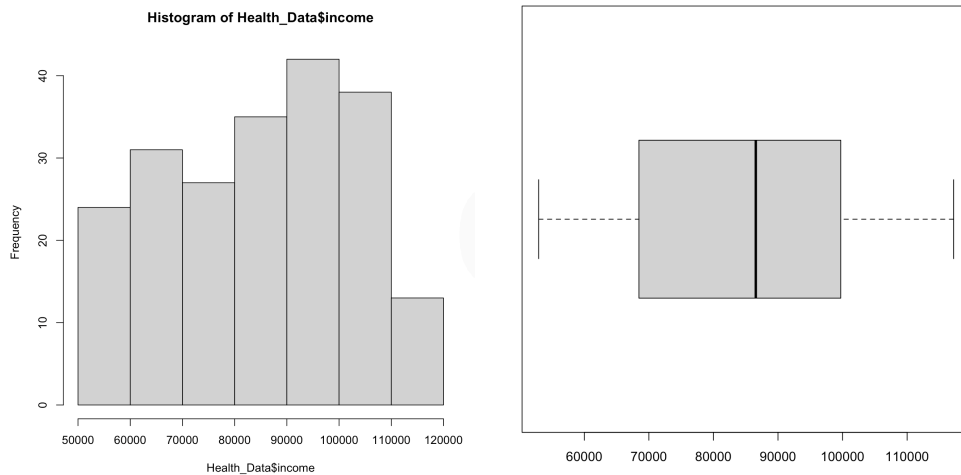
2.1. Central Tendency Measures

Variable	Mean	Median	Mode
sbp	127.7333	123	[1] 120 attr("freq") [1] 12
dbp	82.76667	82	[1] 74 80 82 attr("freq") [1] 13
income	85194.49	86560.5	No mode
age	26.51429	27	[1] 26 attr("freq") [1] 16

2.2. Five-figure summary - Monthly income (income)

Min.	1st Qu.	Median	3rd Qu.	Max.
52933	68636	86560	99696	117210

2.3. Visualisation - Monthly income (income)



The histogram shows a slightly left-skewed distribution. The box plot shows that the median is pretty centred, indicating no strong skewness. The left and right whiskers are of similar length, indicating that the distribution is approximately symmetric. There are also no obvious outliers visible in this group.

3. Hypothesis Testing 1: systolic blood pressure ~ peptic ulcer

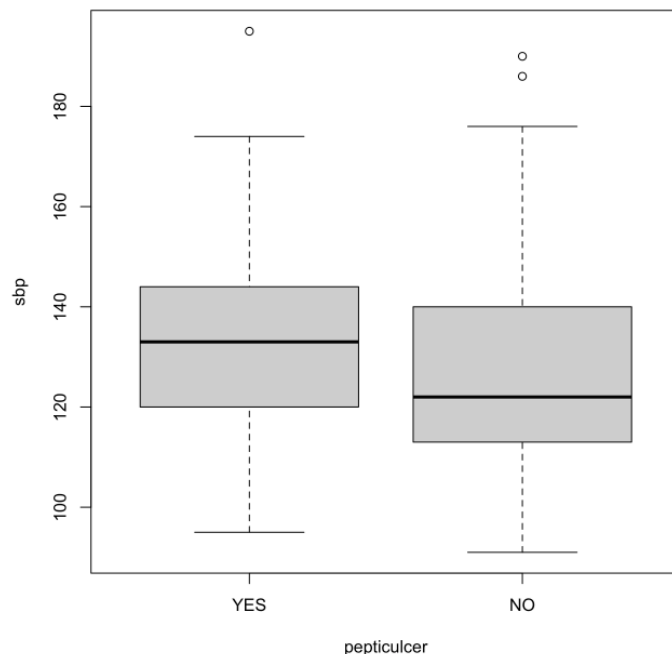
The objective of the research is to determine whether there is an association between systolic blood pressure and the presence or absence of a peptic ulcer.

3.1. Variables Characteristic

Systolic blood pressure is a dependent variable measured, whereas peptic ulcer is an independent variable that may influence and explain the overall outcome.

dependent variable (outcome)	systolic blood pressure	<i>numeric variable, continuous scale</i>
Independent variable (predictor)	peptic ulcer	<i>categorical binary variable with two levels ("Yes"/ "No")</i>

The boxplot indicates a few high-value outliers in both peptic ulcer groups. The whiskers and box widths are similar in length, suggesting that the overall spread of sbp is similar across peptic ulcer groups. There is a slightly higher median in group 1, but overall variability in sbp across the two groups.



3.2. Normality Assumption (sbp)

The Shapiro-Wilk normality test yielded a p-value of 3.345e-06, which is > 0.05 ; hence, we fail to reject H_0 , indicating that 'sbp' follows a normal distribution.

3.3. Hypothesis formulation

H_0 : Mean systolic blood pressure (sbp) is the same for patients with and without peptic ulcer. (**$H_0: \mu_1 = \mu_2$**)

H_1 : Mean systolic blood pressure (sbp) differs between patients with and without peptic ulcer. (**$H_1: \mu_1 \neq \mu_2$**)

3.4. Independent Two-Sample t-Test

An independent Two-Sample t-Test was selected, given that the dependent variable (systolic blood pressure) is numeric and the predictor (peptic ulcer) is a binary categorical binary variable with two independent groups ("Yes"/ "No"). Assessment of

normality indicated that the distribution of systolic blood pressure (sbp) is normal, satisfying the assumptions required for the parametric test.

3.5. Test Results

The p-value of the test is 0.2296, which is greater than the significance level $\alpha = 0.05$, so we fail to reject the null hypothesis ($H_0: \mu_1 = \mu_2$).

Independent Two-Sample t-Test Results	
t = 1.2142	t is the t-test statistic value
df = 57.562	df is the degrees of freedom
p-value = 0.2296	p-value is the significance level of the t-test
conf.int = [-2.889367 11.795703])	conf.int is the confidence interval of the mean at 95%
mean = 131.3171, 126.8639	Sample estimates are the sample mean values.

The results did not reveal a statistically significant difference in mean systolic blood pressure (SBP) between patients with and without a peptic ulcer. Therefore, there is insufficient evidence to conclude that mean SBP differs between the two groups.

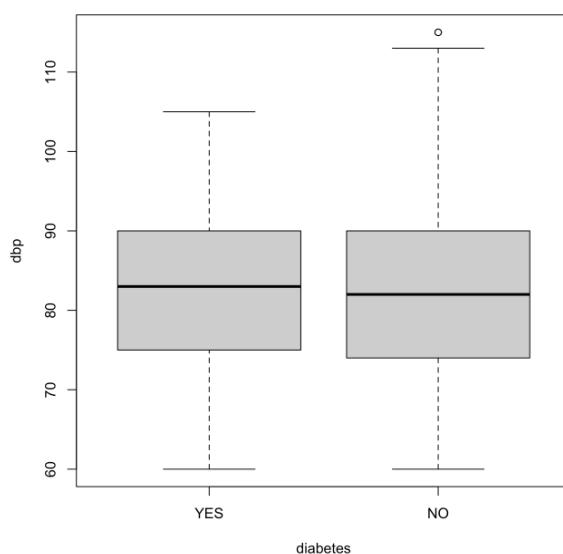
4. Hypothesis Testing 2: diastolic blood pressure ~ diabetic condition

The objective of the hypothesis test is to determine whether there is an association between diastolic blood pressure and diabetic conditions. Diastolic blood pressure is a dependent variable measured, whereas diabetes is an independent variable that may influence the outcome.

dependent variable (outcome)	diastolic blood pressure	<i>numeric variable, continuous scale</i>
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Independent variable (predictor)	diabetes	<i>categorical binary variable with two levels ("Yes"/ "No")</i>
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The boxplot indicates that the median dbp is similar across diabetics and non-diabetics. The whiskers and box widths are similar in length, and there is only one upper outlier in the non-diabetes group, suggesting that the overall spread of dbp is similar across the groups.



4.1. Normality Assumption (diastolic blood pressure)

The Shapiro-Wilk normality test yielded a p-value of 0.0002204, which is smaller than 0.05; hence, we reject H_0 and accept H_1 , indicating that 'diastolic blood pressure' does not follow a normal distribution.

4.2. Hypothesis

H_0 : Median diastolic blood pressure (dbp) is the same among diabetic and non-diabetic participants. (**$H_0: M_1 = M_2$**)

H_1 : Median diastolic blood pressure (dbp) differs among diabetes and non-diabetes. (**$H_1: M_1 \neq M_2$**)

4.3. Mann-Whitney U test

A nonparametric Mann-Whitney U test was selected, given that the dependent variable (diastolic blood pressure) is a continuous numeric variable and the predictor (diabetes) is a binary categorical variable with two independent groups ("Yes" / "No") to compare. Assessment of normality indicated that the distribution of diastolic blood pressure (dbp) violates the assumption of normality, suggesting that a nonparametric test is appropriate.

4.4. Test Results

Mann-Whitney U test Results	
p-value = 0.7999	p-value is the significance level of the t-test

The p-value of the test is 0.7999, which is greater than the significance level $\alpha = 0.05$, so we fail to reject the null hypothesis (**H0: M1 = M2**). The Wilcoxon rank-sum test found no statistically significant difference in diastolic blood pressure between diabetic and non-diabetic groups ($W = 3804.5$, $p\text{-value} = 0.7999$).

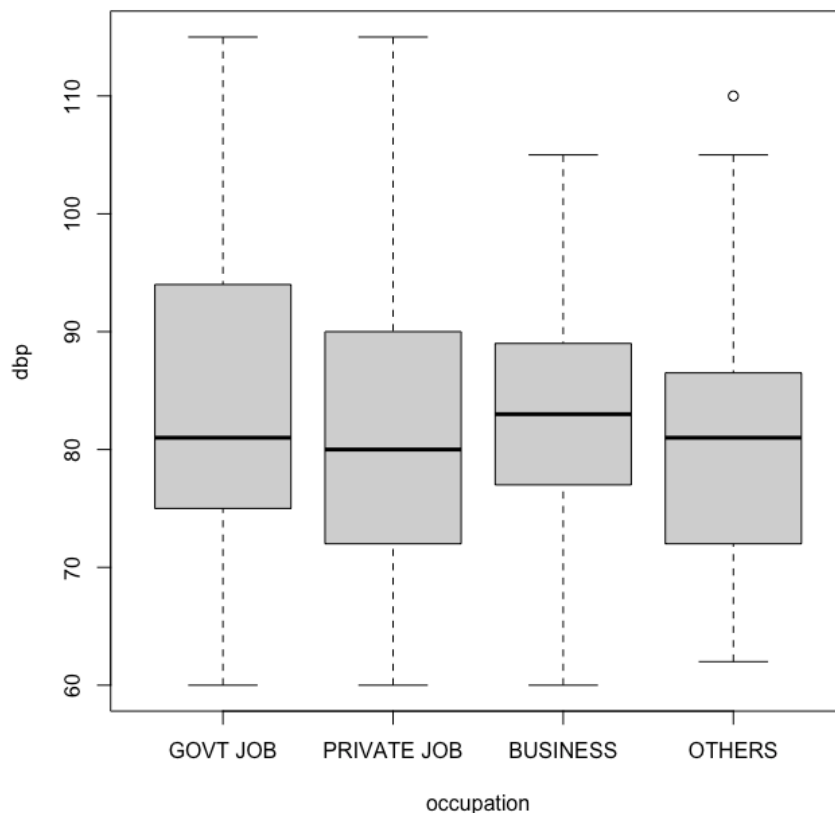
5. Hypothesis Testing 3: systolic blood pressure ~ occupation

The objective of the hypothesis test is to determine whether the systolic blood pressure (sbp) is different across four occupational group ("GOVT JOB", "PRIVATE JOB", "BUSINESS", "OTHERS").

dependent variable (outcome)	systolic blood pressure	<i>numeric variable, continuous scale</i>
Independent variable (predictor)	occupational group	<i>categorical variables with 4 levels ("GOVT JOB", "PRIVATE JOB", "BUSINESS", "OTHERS")</i>

The boxplot indicates that

- The median sbp is approximately similar across 1, 2, and 4 occupational groups, whereas the median of group 3 is slightly higher compared to the other three groups
- All 4 groups have the same box sizes (IQR) and pretty similar whisker lengths, suggesting that the variability across the sbp groups is relatively consistent.
- There is only one outlier in group 4 and no obvious outliers in groups 1,2 and 3.



Visual inspection of the boxplot indicates a similar distribution of dbp across the four occupational groups. The groups exhibit comparable variability, with only one outlier being observed.

5.1. Normality Assumption (systolic blood pressure)

The Shapiro-Wilk normality test yielded a p-value of 3.345e-06, which is > 0.05 ; hence, we fail to reject H_0 , indicating that 'sbp' follows a normal distribution.

5.2. Hypothesis formulation

H_0 : Mean systolic blood pressure (sbp) is the same across the occupational groups
($H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$)

H1: Mean systolic blood pressure (sbp) is at least different in one of the occupational groups (**H1: At least one group differs**)

5.3. One-way ANOVA t-test

A parametric ANOVA t-test was selected, given that the dependent variable (sbp) is a continuous numeric variable and the predictor (occupation) is a categorical variable with four independent groups to compare. Assessment of normality indicated that the distribution of systolic blood pressure (SBP) is approximately normal, suggesting that a parametric test is appropriate.

5.4. Test Results

ANOVA test					
Source	df	SS	MS	F	p-value
Occupation	3	284.80	94.90	0.233	0.873
Residuals	206	83800.27	406.80	–	–

The p-value of the test is 0.873, which is greater than the significance level $\alpha = 0.05$, so we fail to reject the null hypothesis (**H0: $\mu_1 = \mu_2 = \mu_3 = \mu_4$**). The ANOVA test indicated no statistically significant difference in mean systolic blood pressure among the four occupational groups

6. Reflection

This exercise contributed to very valuable learnings and observations:

- Applying a framework for test selection is critical. As Ranganathan (2021) argues, selecting the appropriate test depends on first focusing on descriptive statistics and understanding the research question. Analysing the data's characteristics should precede inferential analysis, during which hypotheses are formulated, and assumptions about normality, outliers, and distribution are validated. This sequence guides the choice of the most suitable test.

- Knowing which variable is being discussed and measured (outcome) and which one may have an effect (predictor) is critical for hypothesis formulation. Moreover, variable types drive test selection. It's important to understand the nature of the variables and how they are stored in R. While working with the health dataset, I realised that variables may be stored as numerical labelled values (num, dbl) while actually representing categories (e.g., Yes/No or "GOVT JOB," "PRIVATE JOB," "BUSINESS," "OTHERS"). This requires careful inspection before analysis, as the statistical meaning might otherwise be changed. Numerical variables can serve as outcomes in a parametric test, while the categorical variables define the groups being compared. Understanding this distinction proved critical for selecting appropriate tests and correctly interpreting results.
- Initially, I focused on applying statistical tests, but the exercise highlighted the importance of data preparation and variable interpretation. Converting labelled variables, such as diabetes and occupation, into factors ensured that R treated them as categorical rather than numerical, leading to more appropriate analyses.
- Assumptions must be checked before testing. Before conducting any test, it is essential to assess assumptions about normality, outliers, and distribution—either through formal tests or visual inspection, such as boxplots. This step highlights that understanding the data first is critical for choosing an appropriate statistical method.
- Hypothesis formulation leads to test selection. I initially focused on formulating a hypothesis and choosing a test, but I have learned that, while the research question stays the same, the hypothesis depends on the test's assumptions and the nature of the variables being compared. This distinction helps decide whether it is more appropriate to compare group means with parametric tests or medians with nonparametric tests. It highlights the importance of choosing statistical methods that align with the data's underlying assumptions.
- Non-significant results do not prove that groups are identical; they simply indicate insufficient evidence to conclude a difference exists. This distinction is key to interpreting hypothesis-testing results. Statistical significance is very different from practical or clinical importance.

- Visuals support interpretation. The box plot provided useful information for identifying potential outliers and comparing variability between groups, generating an initial image and understanding of the data. However, these visuals do not constitute decisive evidence; they must be supported by statistical testing.
- Careful data management is key. Data preparation, reproducibility and proper variable encoding are essential, and debugging is a critical practice that supports them. During the analysis, I discovered that repeatedly converting labelled variables could inadvertently overwrite data and introduce missing values. This highlighted the value of maintaining an untouched copy of the original dataset, performing all variable transformations in a dedicated data preparation section, and structuring the script so that the entire analysis can be reproduced consistently within a trustworthy workflow.
- Finally, although less pronounced in this exercise, as Schober, Bossers and Schwarte (2018) argue, it is important to be mindful that decisions should not be guided solely by p-values. The overall results—including the effect size (how large the effect is) and the confidence interval (how precise the estimate is)—help determine whether the effect is actually meaningful in practice.

This exercise has reinforced the importance of understanding different types of variables, checking statistical assumptions, and then selecting the test that aligns with the research question and data structure. Moreover, it highlights the value of applying a framework that guides data evaluation and preparation, visual exploration, statistical testing, and interpretation when drawing final conclusions.

References

- Schober, P. S., Bossers, S. M. and Schwarte, L. A. (2018) 'Statistical significance versus clinical importance of observed effect sizes: what do P values and confidence intervals really represent?', *Anesthesia & Analgesia*, 126(3), pp. 1068–1072. DOI: 10.1213/ANE.0000000000002798
- Ranganathan, P. (2021) 'An introduction to statistics: choosing the correct statistical test', *Indian Journal of Critical Care Medicine*, 25(Suppl 2), pp. S184–S186. DOI: 10.5005/jp-journals-10071-23815

University of Essex (2026) *Health Survey Dataset* [dataset]. Numerical Analysis
Module. Unpublished teaching dataset.

Appendix:

```
> library(haven)
> Health_Data <- read_sav("Health Data.sav")
> View(Health_Data)
> #summary(Health_Data)
> #sapply(Health_Data, class)
> str(Health_Data)
tibble [210 × 17] (S3: tbl_df/tbl/data.frame)
 $ ID_no      : num [1:210] 1 2 3 4 5 6 7 8 9 10 ...
  .. attr(*, "label")= chr "Id number"
  .. attr(*, "format.spss")= chr "F2.0"
  .. attr(*, "display_width")= int 7
 $ age        : num [1:210] 30 41 14 28 38 26 29 36 20 29 ...
  .. attr(*, "label")= chr "Age"
  .. attr(*, "format.spss")= chr "F8.2"
 $ sex        : chr+lbl [1:210] f, m, m, m, m, m, f, f, f, f, f, m, m, m, f, f, m, m, m, m, m, f, f, f, f, f, f, m, m, m,
 ...
  ..@ label      : chr "Sex: string"
  ..@ format.spss: chr "A1"
  ..@ labels     : Named chr [1:2] "f" "m"
  .. .. attr(*, "names")= chr [1:2] "Female" "Male"
 $ sex_1      : dbl+lbl [1:210] 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1,
 ...
  ..@ label      : chr "Sex: numeric"
  ..@ format.spss : chr "F8.0"
  ..@ display_width: int 10
  ..@ labels     : Named num [1:2] 0 1
  .. .. attr(*, "names")= chr [1:2] "Female" "Male"
 $ religion    : dbl+lbl [1:210] 1, 2, 1, 1, 1, 2, 2, 1, 2, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 2, 2, 1, 2, 1, 1, 1, 1, 1, 1,
 ...
  ..@ label      : chr "Religion"
  ..@ format.spss: chr "F1.0"
  ..@ labels     : Named num [1:3] 1 2 3
  .. .. attr(*, "names")= chr [1:3] "MUSLIM" "HINDU" "Christian"
 $ religion_2  : dbl+lbl [1:210] 1, 4, 1, 1, 1, 4, 2, 1, 4, 1, 1, 1, 1, 1, 1, 4, 1, 1, 1, 4, 2, 1, 4, 1, 1, 1, 1, 1, 1, 1,
 ...
  ..@ label      : chr "Religion 2"
  ..@ format.spss: chr "F8.0"
  ..@ labels     : Named num [1:4] 1 2 3 4
  .. .. attr(*, "names")= chr [1:4] "MUSLIM" "HINDU" "Christian" "BUDDHISM"
```

```

$ occupation : dbl+lbl [1:210] 1, 4, 4, 1, 2, 4, 1, 2, 3, 1, 3, 4, 2, 4, 4, 2, 3, 3, 3, 1, 1, 3, 1, 3, 2, 4, 2, 1, 1,
1, ...
..@ label      : chr "Occupation"
..@ format.spss: chr "F1.0"
..@ labels     : Named num [1:4] 1 2 3 4
..- attr(*, "names")= chr [1:4] "GOVT JOB" "PRIVATE JOB" "BUSINESS" "OTHERS"
$ income      : num [1:210] 79774 70295 100117 105528 71417 ...
..- attr(*, "label")= chr "Monthly income"
..- attr(*, "format.spss")= chr "F6.0"
$ sbp        : num [1:210] 107 105 157 101 101 123 148 100 102 141 ...
..- attr(*, "label")= chr "Systolic BP"
..- attr(*, "format.spss")= chr "F8.0"
$ dbp        : num [1:210] 75 75 90 71 77 86 95 72 80 100 ...
..- attr(*, "label")= chr "Diastolic BP"
..- attr(*, "format.spss")= chr "F8.0"
$ f_history   : dbl+lbl [1:210] 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1,
...
..@ label      : chr "Family history of diabetes"
..@ format.spss : chr "F8.0"
..@ display_width: int 10
..@ labels     : Named num [1:2] 0 1
..- attr(*, "names")= chr [1:2] "No" "Yes"
$ pepticulcer: dbl+lbl [1:210] 2, 1, 2, 2, 2, 1, 2, 1, 2, 1, 2, 1, 2, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
2, ...
..@ label      : chr "Have peptic ulcer"
..@ format.spss: chr "F1.0"
..@ labels     : Named num [1:2] 1 2
..- attr(*, "names")= chr [1:2] "Yes" "No"
$ diabetes    : dbl+lbl [1:210] 2, 1, 2, 2, 1, 1, 1, 2, 2, 2, 1, 2, 2, 1, 2, 2, 2, 1, 2, 1, 1, 2, 2, 1, 2, 2, 2, 2, 1,
...
..@ label      : chr "Have diabetes mellitus"
..@ format.spss: chr "F2.0"
..@ labels     : Named num [1:2] 1 2
..- attr(*, "names")= chr [1:2] "Yes" "No"
$ post_test   : num [1:210] 94 NA NA NA NA NA 100 90.5 96.5 100 ...
..- attr(*, "label")= chr "Post test score"
..- attr(*, "format.spss")= chr "F8.2"
$ pre_test    : num [1:210] 40.5 NA NA NA NA NA 87.5 65.5 51 55.5 ...
..- attr(*, "label")= chr "Pre test score"
..- attr(*, "format.spss")= chr "F8.2"
$ date_ad     : Date[1:210], format: "2020-05-02" "2020-05-02" "2020-05-02" "2020-05-02" ...

```

```

$ date_dis : Date[1:210], format: "2020-05-07" "2020-05-06" "2020-05-08" "2020-05-10" ...
> #Convert variable types
> Health_Data$pepticulcer <- factor(
+ Health_Data$pepticulcer, levels = c(1,2), labels = c("YES", "NO"))
>
> Health_Data$diabetes <- factor(
+ Health_Data$diabetes, levels = c(1,2), labels = c("YES", "NO"))
>
> Health_Data$occupation <- factor(
+ Health_Data$occupation, levels = c(1,2,3,4), labels = c("GOVT JOB", "PRIVATE JOB", "BUSINESS",
"OTHERS"))
> str(Health_Data$occupation)
Factor w/ 4 levels "GOVT JOB","PRIVATE JOB",...: 1 4 4 1 2 4 1 2 3 1 ...
>
> #Check variables
> str(Health_Data)
tibble [210 × 17] (S3: tbl_df/tbl/data.frame)
 $ ID_no      : num [1:210] 1 2 3 4 5 6 7 8 9 10 ...
 ..- attr(*, "label")= chr "Id number"
 ..- attr(*, "format.spss")= chr "F2.0"
 ..- attr(*, "display_width")= int 7
 $ age        : num [1:210] 30 41 14 28 38 26 29 36 20 29 ...
 ..- attr(*, "label")= chr "Age"
 ..- attr(*, "format.spss")= chr "F8.2"
 $ sex        : chr+lbl [1:210] f, m, m, m, m, m, f, f, f, f, f, m, m, m, f, f, m, m, m, m, f, f, f, f, f, f, m, m, m,
 ...
 ..@ label      : chr "Sex: string"
 ..@ format.spss: chr "A1"
 ..@ labels     : Named chr [1:2] "f" "m"
 ..- attr(*, "names")= chr [1:2] "Female" "Male"
 $ sex_1       : dbl+lbl [1:210] 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1,
 ...
 ..@ label      : chr "Sex: numeric"
 ..@ format.spss : chr "F8.0"
 ..@ display_width: int 10
 ..@ labels     : Named num [1:2] 0 1
 ..- attr(*, "names")= chr [1:2] "Female" "Male"
 $ religion    : dbl+lbl [1:210] 1, 2, 1, 1, 1, 2, 2, 1, 2, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 2, 2, 1, 2, 1, 1, 1, 1, 1, 1,
 ...
 ..@ label      : chr "Religion"
 ..@ format.spss: chr "F1.0"

```

```

..@ labels    : Named num [1:3] 1 2 3
.. ..- attr(*, "names")= chr [1:3] "MUSLIM" "HINDU" "Christian"
$ religion_2 : dbl+lbl [1:210] 1, 4, 1, 1, 1, 4, 2, 1, 4, 1, 1, 1, 1, 1, 1, 4, 1, 1, 1, 4, 2, 1, 4, 1, 1, 1, 1, 1,
...
..@ label     : chr "Religion 2"
..@ format.spss: chr "F8.0"
..@ labels    : Named num [1:4] 1 2 3 4
.. ..- attr(*, "names")= chr [1:4] "MUSLIM" "HINDU" "Christian" "BUDDHISM"
$ occupation : Factor w/ 4 levels "GOVT JOB","PRIVATE JOB",...: 1 4 4 1 2 4 1 2 3 1 ...
$ income     : num [1:210] 79774 70295 100117 105528 71417 ...
..- attr(*, "label")= chr "Monthly income"
..- attr(*, "format.spss")= chr "F6.0"
$ sbp       : num [1:210] 107 105 157 101 101 123 148 100 102 141 ...
..- attr(*, "label")= chr "Systolic BP"
..- attr(*, "format.spss")= chr "F8.0"
$ dbp       : num [1:210] 75 75 90 71 77 86 95 72 80 100 ...
..- attr(*, "label")= chr "Diastolic BP"
..- attr(*, "format.spss")= chr "F8.0"
$ f_history  : dbl+lbl [1:210] 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1,
...
..@ label     : chr "Family history of diabetes"
..@ format.spss : chr "F8.0"
..@ display_width: int 10
..@ labels    : Named num [1:2] 0 1
.. ..- attr(*, "names")= chr [1:2] "No" "Yes"
$ pepticulcer: Factor w/ 2 levels "YES","NO": 2 1 2 2 2 1 2 1 2 1 ...
$ diabetes   : Factor w/ 2 levels "YES","NO": 2 1 2 2 1 1 1 2 2 2 ...
$ post_test  : num [1:210] 94 NA NA NA NA NA 100 90.5 96.5 100 ...
..- attr(*, "label")= chr "Post test score"
..- attr(*, "format.spss")= chr "F8.2"
$ pre_test   : num [1:210] 40.5 NA NA NA NA NA 87.5 65.5 51 55.5 ...
..- attr(*, "label")= chr "Pre test score"
..- attr(*, "format.spss")= chr "F8.2"
$ date_ad    : Date[1:210], format: "2020-05-02" "2020-05-02" "2020-05-02" "2020-05-02" ...
$ date_dis   : Date[1:210], format: "2020-05-07" "2020-05-06" "2020-05-08" "2020-05-10" ...
> #summary(Health_Data)
>
> ##
=====
> # Analysis: Mean, Median, Mode for sbp, dbp and income

```

```
> ##
```

```
=====
```

```
>
```

```
> #Data characteristic of Systolic BP (sbp)
```

```
> mean(Health_Data$sbp)
```

```
[1] 127.7333
```

```
> median(Health_Data$sbp)
```

```
[1] 123
```

```
> Mode(Health_Data$sbp)
```

```
[1] 120
```

```
attr("freq")
```

```
[1] 12
```

```
>
```

```
> #Data characteristic - Diastolic BP (dbp)
```

```
> mean(Health_Data$dbp)
```

```
[1] 82.76667
```

```
> median(Health_Data$dbp)
```

```
[1] 82
```

```
> Mode(Health_Data$dbp)
```

```
[1] 74 80 82
```

```
attr("freq")
```

```
[1] 13
```

```
>
```

```
> #Data characteristic of age
```

```
> mean(Health_Data$age)
```

```
[1] 26.51429
```

```
> median(Health_Data$age)
```

```
[1] 27
```

```
> Mode(Health_Data$age)
```

```
[1] 26
```

```
attr("freq")
```

```
[1] 16
```

```
>
```

```
> #Data characteristic - Monthly income (income)
```

```
> mean(Health_Data$income)
```

```
[1] 85194.49
```

```
> median(Health_Data$income)
```

```
[1] 86560.5
```

```
> Mode(Health_Data$income)
```

```
[1] NA
```

```
attr("freq")
```



```

[1] NA
>
> #Five-figure summary - Monthly income (income)
> summary(Health_Data$income)
  Min. 1st Qu.  Median    Mean 3rd Qu.   Max.
52933  68636  86560  85194  99696 117210
>
> #Histogram and Boxplot - Monthly income (income)
> hist(Health_Data$income)  #shape of distribution
> boxplot(Health_Data$income, horizontal = TRUE)
>
> ##
=====
> # Hypothesis Testing 1: systolic blood pressure ~ peptic ulcer
> ##
=====
> #Inspecting data type
> str(Health_Data$sbp)
num [1:210] 107 105 157 101 101 123 148 100 102 141 ...
- attr(*, "label")= chr "Systolic BP"
- attr(*, "format.spss")= chr "F8.0"
> str(Health_Data$pepticulcer)
Factor w/ 2 levels "YES","NO": 2 1 2 2 2 1 2 1 2 1 ...
> boxplot(sbp ~ pepticulcer, data = Health_Data)
>
> ##
=====
> # Normality Testing (sbp)
> ##
=====
> shapiro.test(Health_Data$sbp)

      Shapiro-Wilk normality test

data:  Health_Data$sbp
W = 0.95474, p-value = 3.345e-06

>
> ##
=====
> #Independent Two-Sample t-Test

```

```
> # H0: Mean systolic blood pressure (sbp) is the same for patients with and without peptic ulcer condition.
```

```
> # H1: Mean systolic blood pressure (sbp) differs between patients with and without peptic ulcer condition
```

```
> ##
```

```
=====
```

```
> t.test(sbp~pepticulcer,data = Health_Data)
```

Welch Two Sample t-test

```
data: sbp by pepticulcer
```

```
t = 1.2142, df = 57.562, p-value = 0.2296
```

```
alternative hypothesis: true difference in means between group YES and group NO is not equal to 0
```

```
95 percent confidence interval:
```

```
-2.889367 11.795703
```

```
sample estimates:
```

```
mean in group YES mean in group NO
```

```
131.3171 126.8639
```

```
>
```

```
> ##
```

```
=====
```

```
> # Hypothesis Testing 2: diastolic blood pressure ~ diabetes
```

```
> ##
```

```
=====
```

```
> #Inspecting data type
```

```
> str(Health_Data$dbp)
```

```
num [1:210] 75 75 90 71 77 86 95 72 80 100 ...
```

```
- attr(*, "label")= chr "Diastolic BP"
```

```
- attr(*, "format.spss")= chr "F8.0"
```

```
> str(Health_Data$diabetes)
```

```
Factor w/ 2 levels "YES","NO": 2 1 2 2 1 1 1 2 2 2 ...
```

```
> boxplot(dbp ~ diabetes, data = Health_Data)
```

```
>
```

```
> ##
```

```
=====
```

```
> # Normality Testing (dbp)
```

```
> ##
```

```
=====
```

```
> shapiro.test(Health_Data$dbp)
```

Shapiro-Wilk normality test

```
data: Health_Data$dbp
W = 0.97052, p-value = 0.0002204
```

```
>
> ##
=====
> # Mann-Whitney U test
> # H0: Median diastolic blood pressure (dbp) is the same among diabetic and non-diabetic participants.
> # H1: Median diastolic blood pressure (dbp) differs among diabetes and non-diabetes.
> ##
=====
> wilcox.test(dbp~diabetes,Health_Data)
```

Wilcoxon rank sum test with continuity correction

```
data: dbp by diabetes
W = 3804.5, p-value = 0.7999
alternative hypothesis: true location shift is not equal to 0
```

```
>
> ##
=====
> # Hypothesis Testing 3: systolic blood pressure ~ occupational group
> ##
=====
>
> #Inspecting data type
> str(Health_Data$sbp)
num [1:210] 107 105 157 101 101 123 148 100 102 141 ...
- attr(*, "label")= chr "Systolic BP"
- attr(*, "format.spss")= chr "F8.0"
> str(Health_Data$occupation)
Factor w/ 4 levels "GOVT JOB","PRIVATE JOB",...: 1 4 4 1 2 4 1 2 3 1 ...
> boxplot(dbp ~ occupation, data = Health_Data)
>
> ##
=====
> # Normality Testing (sbp)
> ##
=====
> shapiro.test(Health_Data$sbp)
```

Shapiro-Wilk normality test

```
data: Health_Data$sbp
W = 0.95474, p-value = 3.345e-06
```

```
>
```

```
> ##
```

```
=====
```

```
> ## ANOVA test
```

```
> # H0: Mean systolic blood pressure (sbp) is the same across the occupational groups
```

```
> # H1: Mean systolic blood pressure (sbp) is at least different in one of the occupational groups
```

```
> ##
```

```
=====
```

```
> aov(sbp ~ occupation, data = Health_Data)
```

```
Call:
```

```
  aov(formula = sbp ~ occupation, data = Health_Data)
```

```
Terms:
```

```
      occupation Residuals
```

```
Sum of Squares    284.80 83800.27
```

```
Deg. of Freedom      3    206
```

```
Residual standard error: 20.16922
```

```
Estimated effects may be unbalanced
```

```
> summary(aov(sbp ~ occupation, data = Health_Data))
```

```
      Df Sum Sq Mean Sq F value Pr(>F)
```

```
occupation   3   285    94.9   0.233  0.873
```

```
Residuals 206 83800   406.8
```